

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – STATISTICS I (SUB-STRAND 4.1): FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Welcome back to the Maize Harvest Mystery of Nakuru County! A KALRO officer collected the maize harvests (in 90-kg bags) of 20 smallholder farmers. The county government needs ONE number to represent a typical harvest so it can decide how much subsidised fertiliser to allocate next season. Here is the officer's raw list, in the order she collected it: 12, 7, 21, 12, 4, 10, 28, 12, 15, 9, 12, 31, 6, 10, 20, 12, 14, 8, 24, 11. Use everything you learned in this unit to answer the Driving Question: How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action? Answer each prompt in the space under it. Use full sentences, show any working, and use the 20 Nakuru harvests above as your evidence. Use the correct statistical words (frequency, tally, mean, median, mode, outlier, axis, sector).

Part 1: From chaos to order

At the start of the unit, the KALRO officer's chalkboard list of 20 harvests looked chaotic. (a) Describe how you would build a frequency distribution table from the raw list, including how tally marks are used. (b) Use the list to build your table, then name one pattern that the table lets you see that the raw list hid. (c) Explain one thing the frequency table still cannot tell the county government.

Write your answer here:

Part 2: Three measures, three stories

Use the 20 Nakuru harvests. (a) Explain in your own words how each of the three measures (mean, mode, median) is found, and then find each one for the 20 harvests, showing your working. (b) Explain what your mean tells us using the idea of a 'balance point' or equal redistribution. (c) Compare your mean with your median. Using the idea of outliers, explain why the two values are different.

Write your answer here:

Part 3: Which number should the county use?

The county government must choose ONE number to decide how much subsidised fertiliser to allocate. Write a short Claim–Evidence–Reasoning advisory note to the Nakuru County Director of Agriculture. Choose mean, median or mode and: (a) state your claim, (b) give evidence from the Nakuru data, (c) explain your reasoning, including how outliers affect your choice, and (d) state one limitation or risk of your choice.

Write your answer here:

Part 4: Choosing the right picture

The KALRO officer must present the data to decision-makers. (a) Choose one of the following: a bar graph, a line graph or a pie chart. Describe a question about Kenyan farming or maize that your chosen graph can answer well, and describe how it is constructed (axes, scale, angles or points, as relevant). (b) Name one of the other two graphs and explain why it would be less suitable for the same question. (c) Imagine a bar graph of your frequency table for the 20 harvests, with each harvest value as a category. Which bar would be tallest, and how is it connected to one of the measures of central tendency?

Write your answer here:

Part 5: Answering the Driving Question

Use the whole unit to answer: How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action? Write a short paragraph (5–7 sentences) that follows the journey of the Nakuru maize data from the raw list to a decision. Include at least one organising tool, one measure of central tendency and one graph, and say why the choice of each matters.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Organising data with frequency tables	Describes tally marks (grouped in fives), frequency counts and the building of the table accurately; names a pattern revealed and clearly explains what the table cannot answer.	Describes the table and tally marks correctly and names a pattern; the limitation is stated but not fully explained.	Description of the table or tallying is incomplete or confused; patterns or limitations are missing.
Mean, median and mode: methods and meaning	Explains and correctly calculates all three measures, interprets the mean as a balance point, and explains why the mean is above the median using outliers.	Explains the methods correctly with minor gaps in calculation, in interpreting the mean or in the effect of outliers.	Confuses methods or cannot link the values (13.9, 12, 12) to the data.
Evidence-based recommendation (Claim–Evidence–Reasoning)	Makes a clear claim, supports it with exact Nakuru values, reasons about outliers and names a sensible limitation.	Makes a clear claim with some correct evidence and reasoning; limitation is vague.	Claim is unsupported, evidence is missing or inaccurate, or reasoning is absent.
Choosing and constructing graphs	Matches the graph to a suitable question, describes construction correctly (axes, scale, angles or points), justifies why another graph is less suitable and links the tallest bar to the mode.	Chooses a suitable graph and describes most construction features; the comparison or the mode link is only partly explained.	Graph choice does not fit the question or construction details are missing or wrong.
Answering the Driving Question	Coherent paragraph tracing data from collection to decision, using correct statistical vocabulary and connecting tools to Kenyan decision-making.	Covers most steps with generally correct vocabulary; links to decision-making are brief.	Lists tools without connecting them to the data journey or to the decision.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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